

Task 5: Stochastic Flux Calculations

Skip to Page 9 to skip the lemmas, or skip to Page 18 for results

Even in a uniform distribution, we see movement of particles between bins. I want to figure out a probability distribution that describes these fluctuations in the movement of particles.

The steps we will go through are:

1. Get an expression for the probability that a given particle at some point s crosses a given point x during some timestep Δt .
2. Use this probability to get the mean and variance of all the particles at some point s that will cross x .
3. Use the linearity property of the mean and variance to sum up the probabilities
4. Turn the sum into an integral by letting some step in space Δs go to 0
5. Solve the integral

These steps will go for both, and split at about step 3.

Lemmas: There are some mathematical results that are just algebra and aren't too important to the discussion of the particles. All that work is here.

Lemma 1 This lemma is to show the general formula for this integral:

$$\int_0^\infty t^{2n} e^{-t^2} dt$$

We use integration by parts:

$$\begin{aligned} u &= t^{2n-1} \\ du &= (2n-1)t^{2n-2} dt \\ dv &= te^{-t^2} dt \\ v &= -\frac{1}{2}e^{-t^2} \end{aligned}$$

Which gets us:

$$\begin{aligned} \int_0^\infty t^{2n} e^{-t^2} dt &= \left[-t^{2n-1} \frac{1}{2} e^{-t^2} \right]_0^\infty + \int_0^\infty \frac{1}{2} e^{-t^2} (2n-1) t^{2n-2} dt \\ \int_0^\infty t^{2n} e^{-t^2} dt &= \frac{2n-1}{2} \int_0^\infty t^{2n-2} e^{-t^2} dt \end{aligned}$$

What we have here is a recursive relationship. Define $I(n)$ like so:

$$I(n) = \int_0^\infty t^{2n} e^{-t^2} dt$$

And we can write our recursion as:

$$I(n) = \frac{2n-1}{2} I(n-1)$$

Now just consider the base case, when $n = 0$:

$$I(0) = \int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$$

Now use the recursion to build a definition for any n :

$$\int_0^\infty t^{2n} e^{-t^2} dt = \frac{\sqrt{\pi}}{2} \prod_1^n \frac{2n-1}{2}$$

Lemma 2

This lemma is to show the general formula for this integral:

$$\int_0^\infty t^{2n-1}(1 - \operatorname{erf}(t)) dt$$

Begin with integration by parts:

$$\begin{aligned} u &= 1 - \operatorname{erf}(t) \\ du &= -\frac{2}{\sqrt{\pi}}e^{-t^2} dt \\ dv &= t^{2n-1} dt \\ v &= \frac{t^{2n}}{2n} \end{aligned}$$

Which gets us:

$$\begin{aligned} \int_0^\infty t^{2n-1}(1 - \operatorname{erf}(t)) dt &= \left[(1 - \operatorname{erf}(t)) \frac{t^{2n}}{2n} \right]_0^\infty + \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{t^{2n}}{2n} e^{-t^2} dt \\ \int_0^\infty t^{2n-1}(1 - \operatorname{erf}(t)) dt &= \frac{1}{n\sqrt{\pi}} \int_0^\infty t^{2n} e^{-t^2} dt \end{aligned}$$

Use **Lemma 1**:

$$\begin{aligned} \int_0^\infty t^{2n-1}(1 - \operatorname{erf}(t)) dt &= \frac{1}{n\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \prod_1^n \frac{2n-1}{2} \\ \int_0^\infty t^{2n-1}(1 - \operatorname{erf}(t)) dt &= \frac{1}{2^{n+1}n} \prod_1^n (2n-1) \end{aligned}$$

Lemma 3

This lemma is to get the general formula for the following integral:

$$F(x, n) = 2 \int_0^{\sqrt{x}} t^{2n+1} e^{-t^2} dt$$

Substitute $s = t^2$, $ds = 2t dt$:

$$F(x, n) = \int_0^x s^n e^{-s} ds$$

Integration by parts:

$$\begin{aligned} u &= s^n \\ du &= ns^{n-1} \\ dv &= e^{-s} ds \\ v &= -e^{-s} \end{aligned}$$

Gets us:

$$\begin{aligned} F(x, n) &= [-s^n e^{-s}]_0^x + n \int_0^x s^{n-1} e^{-s} ds \\ F(x, n) &= -x^n e^{-x} + nF(x, n-1) \end{aligned}$$

We have to now expand the recursive definition. First, the base case. Consider $n = 0$:

$$\begin{aligned} F(x, 0) &= \int_0^x e^{-s} ds \\ F(x, 0) &= 1 - e^{-x} \end{aligned}$$

Now let's unpack the recursion:

$$\begin{aligned} F(x, n) &= -x^n e^{-x} + nF(x, n-1) \\ F(x, n) &= -x^n e^{-x} + n(-x^{n-1} e^{-x} + (n-1)F(x, n-2)) \\ F(x, n) &= -x^n e^{-x} - nx^{n-1} e^{-x} + n(n-1)x^{n-2} + \dots + n!F(x, 0) \\ F(x, n) &= -x^n e^{-x} - nx^{n-1} e^{-x} + n(n-1)x^{n-2} + \dots + n! - n!e^{-x} \\ F(x, n) &= n! - e^{-x} n! \sum_{i=0}^n \frac{x^i}{i!} \\ F(x, n) &= n! \left(1 - e^{-x} \sum_{i=0}^n \frac{x^i}{i!} \right) \end{aligned}$$

Lemma 4

This lemma is to get the value of the following integral:

$$\int_0^\infty t^{2n}(1 - \operatorname{erf}(t)^2) dt$$

First, integrate by parts:

$$\begin{aligned} u &= 1 - \operatorname{erf}(t)^2 \\ du &= -\frac{4}{\sqrt{\pi}}e^{-t^2}\operatorname{erf}(t) dt \\ dv &= t^{2n} dt \\ v &= \frac{t^{2n+1}}{2n+1} \end{aligned}$$

Getting us:

$$\begin{aligned} \int_0^\infty t^{2n}(1 - \operatorname{erf}(t)^2) dt &= \left[(1 - \operatorname{erf}(t)^2) \frac{t^{2n+1}}{2n+1} \right]_0^\infty + \frac{4}{(2n+1)\sqrt{\pi}} \int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt \\ \int_0^\infty t^{2n}(1 - \operatorname{erf}(t)^2) dt &= \frac{4}{(2n+1)\sqrt{\pi}} \int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt \end{aligned}$$

Integration by parts again. We get v using **Lemma 3**:

$$\begin{aligned} u &= \operatorname{erf}(t) \\ du &= \frac{2}{\sqrt{\pi}}e^{-t^2} dt \\ dv &= t^{2n+1}e^{-t^2} dt \\ v &= \int_0^x t^{2n+1}e^{-t^2} dt = \frac{1}{2}F(x^2, n) = \frac{1}{2}n! \left(1 - e^{-x} \sum_{i=0}^n \frac{x^i}{i!} \right) \end{aligned}$$

Gets us:

$$\begin{aligned} &\int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt = \\ &\frac{1}{2} \left[\operatorname{erf}(t) n! \left(1 - e^{-x^2} \sum_{i=0}^n \frac{x^{2i}}{i!} \right) \right]_0^\infty - \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-t^2} n! \left(1 - e^{-t^2} \sum_{i=0}^n \frac{t^{2i}}{i!} \right) dt \\ &\int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt = \frac{1}{2}n! - \frac{1}{\sqrt{\pi}}n! \int_0^\infty e^{-t^2} \left(1 - e^{-t^2} \sum_{i=0}^n \frac{t^{2i}}{i!} \right) dt \\ &\int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt = \frac{1}{2}n! - \frac{1}{\sqrt{\pi}}n! \left(\frac{\sqrt{\pi}}{2} - \int_0^\infty e^{-2t^2} \sum_{i=0}^n \frac{t^{2i}}{i!} dt \right) \\ &\int_0^\infty t^{2n+1}e^{-t^2}\operatorname{erf}(t) dt = \frac{1}{2}n! - \frac{1}{\sqrt{\pi}}n! \left(\frac{\sqrt{\pi}}{2} - \sum_{i=0}^n \frac{1}{i!} \int_0^\infty t^{2i}e^{-2t^2} dt \right) \end{aligned}$$

Substitute $u = t\sqrt{2}$, $du = dt\sqrt{2}$:

$$\int_0^\infty t^{2n+1} e^{-t^2} \operatorname{erf}(t) dt = \frac{1}{2}n! - \frac{1}{\sqrt{\pi}}n! \left(\frac{\sqrt{\pi}}{2} + \sum_{i=0}^n \frac{1}{i! \cdot 2^{i+1/2}} \int_0^\infty u^{2i} e^{-u^2} du \right)$$

Substitute using **Lemma 1**:

$$\begin{aligned} \int_0^\infty t^{2n} e^{-t^2} dt &= \frac{\sqrt{\pi}}{2} \prod_1^n \frac{2n-1}{2} \\ \int_0^\infty t^{2n+1} e^{-t^2} \operatorname{erf}(t) dt &= \frac{1}{2}n! - \frac{1}{\sqrt{\pi}}n! \left(\frac{\sqrt{\pi}}{2} - \sum_{i=0}^n \frac{1}{i! \cdot 2^{i+1/2}} \frac{\sqrt{\pi}}{2} \prod_1^i \frac{2i-1}{2} \right) \\ \int_0^\infty t^{2n+1} e^{-t^2} \operatorname{erf}(t) dt &= \frac{1}{2}n! \sum_{i=0}^n \frac{1}{i! \cdot 2^{i+1/2}} \prod_1^i \frac{2i-1}{2} \\ \int_0^\infty t^{2n+1} e^{-t^2} \operatorname{erf}(t) dt &= \frac{1}{2\sqrt{2}}n! \sum_{i=0}^n \frac{1}{i! \cdot 2^{2i}} \prod_1^i (2i-1) \end{aligned}$$

Now we solve the original problem:

$$\begin{aligned} \int_0^\infty t^{2n} (1 - \operatorname{erf}(t)^2) dt &= \frac{4}{(2n+1)\sqrt{\pi}} \int_0^\infty t^{2n+1} e^{-t^2} \operatorname{erf}(t) dt \\ \int_0^\infty t^{2n} (1 - \operatorname{erf}(t)^2) dt &= \frac{\sqrt{2}}{(2n+1)\sqrt{\pi}} n! \sum_{i=0}^n \frac{1}{i! \cdot 2^{2i}} \prod_1^i (2i-1) \end{aligned}$$

Lemma 5:

This lemma is the Taylor expansion of the following expression:

$$\frac{f(x+h) - f(x-h)}{2h}$$

Let's expand this by substituting the Taylor polynomials for each f . Note that this assumes f is analytical

$$\begin{aligned} f(x+h) &= \sum_{i=0}^{\infty} \frac{h^i}{i!} f^{(i)}(x) \\ f(x-h) &= \sum_{i=0}^{\infty} \frac{(-1)^i h^i}{i!} f^{(i)}(x) \\ f(x+h) - f(x-h) &= \sum_{i=0}^{\infty} \frac{(1 - (-1)^i) h^i}{i!} f^{(i)}(x) \end{aligned}$$

Note that for all even numbers, $1 - (-1)^i = 0$, and for all odd numbers, it's two. So this reduces:

$$\begin{aligned} f(x+h) - f(x-h) &= 2 \sum_{i=0}^{\infty} \frac{h^{2i+1}}{(2i+1)!} f^{(2i+1)}(x) \\ \frac{f(x+h) - f(x-h)}{2h} &= \sum_{i=0}^{\infty} \frac{h^{2i}}{(2i+1)!} f^{(2i+1)}(x) \end{aligned}$$

Lemma 6

Lemma 6 is the Taylor expansion of the following expression:

$$f(x - h) + f(x + h)$$

We write the correct expansions out and simplify.

$$\begin{aligned} f(x + h) &= \sum_{i=0}^{\infty} \frac{h^i}{i!} f^{(i)}(x) \\ f(x - h) &= \sum_{i=0}^{\infty} \frac{(-1)^i h^i}{i!} f^{(i)}(x) \\ f(x + h) + f(x - h) &= \sum_{i=0}^{\infty} \frac{(1 + (-1)^i) h^i}{i!} f^{(i)}(x) \end{aligned}$$

This time, only the even terms are left:

$$f(x + h) + f(x - h) = 2 \sum_{i=0}^{\infty} \frac{h^{2i}}{(2i)!} f^{(2i)}(x)$$

I: Particle Probability Distributions

Our particles move according to a random walk. The probability distribution of the possible displacements of our particle is:

$$f(\Delta x) = \frac{1}{2\sqrt{\pi D \Delta t}} e^{-\frac{\Delta x^2}{4D \Delta t}} \quad (1)$$

Consider a particle located at s , to the left of some point x , meaning $s < x$. This particle can cross x going to the right as long as its Δx is greater than the distance between x and s , or $x - s$. To get the probability of this event, it's simply the probability that $\Delta x > x - s$. We can call this p^+ , for the probability that a particle crosses rightwards:

$$p^+ = \frac{1}{2\sqrt{\pi D \Delta t}} \int_{x-s}^{\infty} e^{-\frac{\Delta x^2}{4D \Delta t}} d(\Delta x) \quad (2)$$

$$p^+ = \frac{2\sqrt{D \Delta t}}{2\sqrt{\pi D \Delta t}} \int_{\frac{x-s}{2\sqrt{D \Delta t}}}^{\infty} e^{-u^2} du \quad (3)$$

$$p^+ = \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{x-s}{2\sqrt{D \Delta t}} \right) \right) \quad (4)$$

Then we have the probability that a particle on the right crosses leftward. This is almost the same as before, except that $\Delta x < x - s$, since s is greater than x now. Since the distribution of Δx is even, we can replace Δx with $-\Delta x$ and the probabilities stay the same, so the probability that $\Delta x < x - s$ is the same as $\Delta x > s - x$. This is the same as p^+ except with $x - s$ replaced with $s - x$:

$$p^- = \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{s-x}{2\sqrt{D \Delta t}} \right) \right) \quad (5)$$

With this we move onto the next step.

II: Binomial Distribution

For a given point s with particles on it, the distribution of the number of particles that start at s which also cross x is binomial. This is because the probability that any one of them crosses is a constant, and they are independent of each other. So, at any given point s , the number of particles who will cross some point x , has mean (μ_s) and variance (σ_s^2) given by:

$$\mu_s = Np \quad (6)$$

$$\sigma_s^2 = Np(1 - p) \quad (7)$$

(Note that just p is used here since this applies both to particles going left and particles going right).

III: Combined Distribution

To get the overall mean and variance of the amount of particles expected to go both left and right, we must use the properties of mean and variance. Namely:

$$E\left[\sum X\right] = \sum E[X] \quad (8)$$

$$\text{Var}\left(\sum X\right) = \sum \text{Var}(X) \quad (9)$$

So what we can do is sum the mean and variance of all the particles going right, and all the particles going left, and then add those together.

Consider a bin of size Δs in the x direction with its left edge on the point $x = s$. If Δs is sufficiently small enough, we can estimate the x coordinate of all of the bins inside of it to be simply s , thus they have the binomial distribution. The number of particles in a given bin is approximated by the length of the bin times the number density of the particles. That gives us:

$$\mu_s = n(s, t)p\Delta s$$

$$\sigma_s^2 = n(s, t)p(1 - p)\Delta s$$

Though keep in mind we would like to figure out in all how many particles are going left and right. So, we have to take the sum of all bins of size Δs on either side:

$$\mu = \sum n(s, t)p\Delta s$$

$$\sigma^2 = \sum n(s, t)p(1 - p)\Delta s$$

And as we let $\Delta s \rightarrow 0$, our sum becomes an integral:

$$\mu = \int n(s, t)p ds \quad (10)$$

$$\sigma^2 = \int n(s, t)p(1 - p) ds \quad (11)$$

IV: Setting up the Integrals

With (10) and (11), we are almost there. However, note that the probability function p is different depending on which side of x we are on. Let's come up with integrals representing the mean and variance of particles going both left and right:

$$\mu^- = \int_x^\infty n(s, t) p^- ds \quad (12)$$

$$\mu^+ = \int_{-\infty}^x n(s, t) p^+ ds \quad (13)$$

$$\sigma_-^2 = \int_x^\infty n(s, t) p^- (1 - p^-) ds \quad (14)$$

$$\sigma_+^2 = \int_{-\infty}^x n(s, t) p^+ (1 - p^+) ds \quad (15)$$

$$(16)$$

(Note: The reason we can take the bound going to infinity is because $n(x, t)$ is technically defined for all real n as being periodic on L . We could split this up into multiple equivalent integrals using the property that $n(x + L, t) = n(x, t)$, but that isn't necessary.)

To get the mean of particles going both left and right, we have to add μ_+ to $-\mu_-$. This is because μ_- tracks particles going left, which we consider negative.

To get the variance, we do just add σ_-^2 and σ_+^2 . This is because when adding variances, direction doesn't matter. We get the following expressions for both.

$$\mu = \int_{-\infty}^x n(s, t) p^+ ds - \int_x^\infty n(s, t) p^- ds \quad (17)$$

$$\sigma^2 = \int_{-\infty}^x n(s, t) p^+ (1 - p^+) ds + \int_x^\infty n(s, t) p^- (1 - p^-) ds \quad (18)$$

V. Solving the Integrals: μ

First, substitute in our definitions for p^+ and p^- :

$$\mu = \int_{-\infty}^x n(s, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{x-s}{2\sqrt{D\Delta t}} \right) \right) ds - \int_x^{\infty} n(s, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{s-x}{2\sqrt{D\Delta t}} \right) \right) ds \quad (19)$$

Apply a transformation to make the integral bounds the same: $u = x - s$, $du = -ds$ for the first one, $u = s - x$, $du = ds$ for the second:

$$\mu = - \int_{\infty}^0 n(x-u, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du - \int_0^{\infty} n(x+u, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (20)$$

$$\mu = \int_0^{\infty} n(x-u, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du - \int_0^{\infty} n(x+u, t) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (21)$$

$$\mu = \int_0^{\infty} (n(x-u, t) - n(x+u, t)) \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (22)$$

$$\mu = \int_0^{\infty} \frac{n(x-u, t) - n(x+u, t)}{2u} \cdot u \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (23)$$

Now let's apply the series expansion from Lemma 5. Note that $n^{(a)}(x, t)$ denotes x derivatives:

$$\mu = - \int_0^{\infty} \sum_{i=0}^{\infty} \frac{u^{2i}}{(2i+1)!} n^{(2i+1)}(x, t) \cdot u \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (24)$$

$$\mu = - \sum_{i=0}^{\infty} \frac{1}{(2i+1)!} n^{(2i+1)}(x, t) \int_0^{\infty} u^{2i+1} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right) \right) du \quad (25)$$

Substitute $2\sqrt{D\Delta t}z = u$, $2\sqrt{D\Delta t}dz = du$:

$$\mu = - \sum_{i=0}^{\infty} \frac{1}{(2i+1)!} n^{(2i+1)}(x, t) \int_0^{\infty} (2\sqrt{D\Delta t}z)^{2i+1} (1 - \operatorname{erf}(z)) 2\sqrt{D\Delta t}dz \quad (26)$$

$$\mu = - \sum_{i=0}^{\infty} \frac{(4D\Delta t)^{i+1}}{(2i+1)!} n^{(2i+1)}(x, t) \int_0^{\infty} z^{2i+1} (1 - \operatorname{erf}(z)) dz \quad (27)$$

We can use the result of **Lemma 2**:

$$\int_0^{\infty} t^{2n-1} (1 - \operatorname{erf}(t)) dt = \frac{1}{2^{n+1}n} \prod_{1}^n (2n-1) \quad (28)$$

$$(29)$$

Where $i + 1$ takes the job of n :

$$\mu = - \sum_{i=0}^{\infty} \frac{(4D\Delta t)^{i+1}}{(2i+1)!} n^{(2i+1)}(x, t) \frac{1}{2^{i+2}(i+1)} \prod_{j=1}^{i+1} (2j-1) \quad (30)$$

$$\mu = - \sum_{i=0}^{\infty} \frac{(4D\Delta t)^{i+1}}{2^{i+2}(2i+1)!} n^{(2i+1)}(x, t) \frac{1}{(i+1)} \prod_{j=1}^{i+1} (2j-1) \quad (31)$$

$$\mu = - \sum_{i=0}^{\infty} \frac{(2D\Delta t)^{i+1}}{2(2i+1)!} n^{(2i+1)}(x, t) \frac{1}{(i+1)} \prod_{j=1}^{i+1} (2j-1) \quad (32)$$

For good measure, we also divide by Δt to get number of particles per second (correct units):

$$\mu = -D \sum_{i=0}^{\infty} \frac{(2D\Delta t)^i}{(2i+1)!} n^{(2i+1)}(x, t) \frac{1}{(i+1)} \prod_{j=1}^{i+1} (2j-1) \quad (33)$$

Let's separate things out. Firstly, everything that's not D , Δt , or n will multiply out to a number, so our formula can be simplified:

$$\mu = -D \sum_{i=0}^{\infty} a_i (D\Delta t)^i n^{(2i+1)}(x, t) \quad (34)$$

$$a_i = \frac{2^i}{(2i+1)!(i+1)} \prod_{j=1}^{i+1} (2j-1) \quad (35)$$

Note that $(2i+1)!$ can be thought of as the product of the first $i+1$ odd numbers and the first i even numbers:

$$a_i = \frac{2^i}{(i+1)} \prod_{j=1}^i \frac{1}{2j} \prod_{j=1}^{i+1} \frac{1}{2j-1} \prod_{j=1}^{i+1} (2j-1) \quad (36)$$

$$a_i = \frac{2^i}{(i+1)} \prod_{j=1}^i \frac{1}{2j} \quad (37)$$

$$a_i = \frac{2^i}{2^i(i+1)} \prod_{j=1}^i \frac{1}{j} \quad (38)$$

$$a_i = \frac{1}{(i+1)} \frac{1}{i!} \quad (39)$$

$$a_i = \frac{1}{(i+1)!} \quad (40)$$

Rewriting the sum:

$$\mu = -D \sum_{i=0}^{\infty} \frac{(D\Delta t)^i}{(i+1)!} n^{(2i+1)}(x, t)$$

With this we can expand out and look at the first few terms, to see if they match our expectations:

$$\mu = -D \frac{\partial n}{\partial x} - \frac{1}{2} D^2 \Delta t \frac{\partial^3 n}{\partial x^3} - \frac{1}{6} D^3 \Delta t^2 \frac{\partial^5 n}{\partial x^5} + \dots \quad (41)$$

Strikingly, the first term is exactly what is computed through analyzing the differential equations, but now we see we have extra terms that decay in proportion to Δt and D . So long as we have sufficiently small values for both, this will work out.

V. Solving the Integrals: Variance

Here's the expression for the variance again:

$$\sigma^2 = \int_{-\infty}^x n(s, t) p^+ (1 - p^+) ds + \int_x^{\infty} n(s, t) p^- (1 - p^-) ds \quad (42)$$

We can substitute the known expressions for p^+ and p^- :

$$p^+ = \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{x - s}{2\sqrt{D\Delta t}} \right) \right) \quad (43)$$

$$p^+ (1 - p^+) = \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{x - s}{2\sqrt{D\Delta t}} \right) \right) \left(1 + \operatorname{erf} \left(\frac{x - s}{2\sqrt{D\Delta t}} \right) \right) \quad (44)$$

$$p^+ (1 - p^+) = \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{x - s}{2\sqrt{D\Delta t}} \right) \right)^2 \quad (45)$$

$$p^- (1 - p^-) = \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{s - x}{2\sqrt{D\Delta t}} \right) \right)^2 \quad (46)$$

$$\sigma^2 = \int_{-\infty}^x n(s, t) \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{x - s}{2\sqrt{D\Delta t}} \right) \right)^2 ds + \int_x^{\infty} n(s, t) \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{s - x}{2\sqrt{D\Delta t}} \right) \right)^2 ds \quad (47)$$

Now we transform. For the left integral, $u = x - s$, $du = -ds$. For the right

integral, $u = s - x$, $du = ds$:

$$\sigma^2 = - \int_{-\infty}^0 n(x-u, t) \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) du + \int_0^{\infty} n(x+u, t) \frac{1}{4} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) du \quad (48)$$

$$\sigma^2 = \frac{1}{4} \int_0^{\infty} (n(x-u, t) + n(x+u, t)) \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) du \quad (49)$$

We can bring in the series from Lemma 6:

$$f(x+h) + f(x-h) = 2 \sum_{i=0}^{\infty} \frac{h^{2i}}{(2i)!} f^{(2i)}(x) \quad (50)$$

$$\sigma^2 = \frac{1}{4} \int_0^{\infty} 2 \sum_{i=0}^{\infty} \frac{u^{2i}}{(2i)!} n^{(2i)}(x, t) \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) du \quad (51)$$

$$\sigma^2 = \frac{1}{2} \sum_{i=0}^{\infty} \frac{1}{(2i)!} n^{(2i)}(x, t) \int_0^{\infty} u^{2i} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) du \quad (52)$$

Substitute $u = 2z\sqrt{D\Delta t}$:

$$\sigma^2 = \frac{1}{2} \sum_{i=0}^{\infty} \frac{1}{(2i)!} n^{(2i)}(x, t) \int_0^{\infty} u^{2i} \left(1 - \operatorname{erf} \left(\frac{u}{2\sqrt{D\Delta t}} \right)^2 \right) (2\sqrt{D\Delta t})^{2i+1} dz \quad (53)$$

$$\sigma^2 = \frac{1}{2} \sum_{i=0}^{\infty} \frac{(2\sqrt{D\Delta t})^{2i+1}}{(2i)!} n^{(2i)}(x, t) \int_0^{\infty} z^{2i} (1 - \operatorname{erf}(z)^2) dz \quad (54)$$

Use Lemma 4 :

$$\int_0^{\infty} t^{2n} (1 - \operatorname{erf}(t)^2) dt = \frac{\sqrt{2}}{(2n+1)\sqrt{\pi}} n! \sum_{i=0}^n \frac{1}{i! \cdot 2^{2i}} \prod_{k=1}^i (2k-1) \quad (55)$$

$$\sigma^2 = \frac{1}{2} \sum_{i=0}^{\infty} \frac{(2\sqrt{D\Delta t})^{2i+1}}{(2i)!} n^{(2i)}(x, t) \frac{\sqrt{2}}{(2i+1)\sqrt{\pi}} i! \sum_{j=0}^i \frac{1}{j! \cdot 2^{2j}} \prod_{k=1}^j (2k-1) \quad (56)$$

Again this is simply an infinite sum of derivatives of n . Let's separate the terms that don't have to do with D or Δt and simplify:

$$\sigma^2 = \sqrt{D\Delta t} \sum_{i=0}^{\infty} a_i (D\Delta t)^i n^{(2i)}(x, t) \quad (57)$$

$$a_i = \frac{2^{2i}}{(2i)!} \frac{\sqrt{2}}{(2i+1)\sqrt{\pi}} i! \sum_{j=0}^i \frac{1}{j! \cdot 2^{2j}} \prod_{k=1}^j (2k-1) \quad (58)$$

Let's simplify the terms in the sum first:

$$\frac{1}{j! \cdot 2^{2j}} \prod_{k=1}^j (2k-1) \quad (59)$$

$$\frac{1}{2^j} \left(\prod_{k=1}^j \frac{1}{2k} \right) \prod_{k=1}^j (2k-1) \quad (60)$$

$$\frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \quad (61)$$

Unfortunately it is hard to compute this sum in general. Let's keep it in this form and continue:

$$a_i = \frac{2^{2i}}{(2i)!} \frac{\sqrt{2}}{(2i+1)\sqrt{\pi}} i! \sum_{j=0}^i \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \quad (62)$$

$$a_i = \frac{2^{2i} i!}{(2i+1)!} \sqrt{\frac{2}{\pi}} \sum_{j=0}^i \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \quad (63)$$

$$(64)$$

Now we can try to do analysis. Let's consider how this term grows as $i \rightarrow \infty$. The first part of the term, with the factorials, shrinks with $i!$. The $\sqrt{2/\pi}$ is a constant. Inside the summation is a product, which shrinks slowly, and an exponential, which shrinks exponentially. The dominant term is the factorial term, so this as a whole shrinks with $i!$. This is useful, as it means only including the first few terms works fine as an approximation.

Let's compute some terms:

i = 0:

$$\begin{aligned} a_0 &= \frac{2^0 0!}{(1)!} \sqrt{\frac{2}{\pi}} \sum_{j=0}^0 \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \\ a_0 &= \sqrt{\frac{2}{\pi}} \left(\frac{1}{2^0} \right) \\ a_0 &= \sqrt{\frac{2}{\pi}} \end{aligned}$$

i = 1:

$$\begin{aligned}
a_1 &= \frac{2^2 1!}{(2+1)!} \sqrt{\frac{2}{\pi}} \sum_{j=0}^1 \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \\
a_1 &= \frac{4}{6} \sqrt{\frac{2}{\pi}} \left(1 + \frac{1}{2} \left(\frac{1}{2} \right) \right) \\
a_1 &= \frac{4}{6} \sqrt{\frac{2}{\pi}} \left(\frac{5}{4} \right) \\
a_1 &= \frac{5}{3\sqrt{2\pi}}
\end{aligned}$$

i = 2:

$$\begin{aligned}
a_2 &= \frac{2^4 2!}{(4+1)!} \sqrt{\frac{2}{\pi}} \sum_{j=0}^2 \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \\
a_2 &= \frac{2^4}{5(4)(3)} \sqrt{\frac{2}{\pi}} \left(1 + \frac{1}{2} \left(\frac{1}{2} \right) + \frac{1}{4} \left(\frac{1}{2} \cdot \frac{3}{4} \right) \right) \\
a_2 &= \frac{2^4}{5(4)(3)} \sqrt{\frac{2}{\pi}} \left(1 + \frac{1}{4} + \frac{3}{32} \right) \\
a_2 &= \frac{4}{15} \sqrt{\frac{2}{\pi}} \left(\frac{3+8+32}{32} \right) \\
a_2 &= \frac{4}{15} \sqrt{\frac{2}{\pi}} \left(\frac{43}{32} \right) \\
a_2 &= \frac{43}{120} \sqrt{\frac{2}{\pi}}
\end{aligned}$$

For the variance in the number of particles per second, we must divide everything by Δt^2 . We finally arrive at:

$$\sigma^2 = \sqrt{\frac{2D}{\pi \Delta t^3}} n(x, t) + \frac{5}{6} \sqrt{\frac{2D^3}{\pi \Delta t}} \frac{\partial^2 n}{\partial x^2} + \frac{43}{120} \sqrt{\frac{2D^5 \Delta t}{\pi}} \frac{\partial^4 n}{\partial x^4} + \dots \quad (65)$$

This formula almost works. There's something curious about it: when testing with actual data (with a constant n , so only the first term is used), it is off by a factor of exactly $\sqrt{2}$ and I don't know why this could be.

Summary:

We found these series expansions for μ and σ^2 :

$$\mu = -D \sum_{i=0}^{\infty} \frac{(D\Delta t)^i}{(i+1)!} \frac{\partial^{2i+1} n}{\partial x^{2i+1}} \quad (66)$$

$$\sigma^2 = \sqrt{\frac{2D}{\pi\Delta t^3}} \sum_0^{\infty} \left((D\Delta t)^i \frac{\partial^{2i} n}{\partial x^{2i}} \frac{2^{2i} i!}{(2i+1)!} \sum_{j=0}^i \frac{1}{2^j} \prod_{k=1}^j \frac{2k-1}{2k} \right) \quad (67)$$

First few terms:

$$\mu = -D \frac{\partial n}{\partial x} - \frac{1}{2} D^2 \Delta t \frac{\partial^3 n}{\partial x^3} - \frac{1}{6} D^3 \Delta t^2 \frac{\partial^5 n}{\partial x^5} + \dots \quad (68)$$

$$\sigma^2 = \sqrt{\frac{2D}{\pi\Delta t^3}} n(x, t) + \frac{5}{6} \sqrt{\frac{2D^3}{\pi\Delta t}} \frac{\partial^2 n}{\partial x^2} + \frac{43}{120} \sqrt{\frac{2D^5 \Delta t}{\pi}} \frac{\partial^4 n}{\partial x^4} + \dots \quad (69)$$

They appear to be accurate for μ , but inaccurate for σ by a factor of $\sqrt{2}$.